Ex No: 2b

IMPLEMENTATION OF UNIFICATION AND RESOLUTION ALGORITHM

Aim:

To implement unification and resolution algorithm using python.

Procedure:

Program:

```
# Unification and Resolution in Python

def unify(var1, var2, subst):
    """Unify two variables or terms with substitution."""
    if subst is None:
        return None
    elif var1 == var2:
        return subst
    elif isinstance(var1, str) and var1.islower():
        return unify_var(var1, var2, subst)
    elif isinstance(var2, str) and var2.islower():
        return unify_var(var2, var1, subst)
    elif isinstance(var1, list) and isinstance(var2, list) and len(var1) == len(var2):
        return unify(var1[1:], var2[1:], unify(var1[0], var2[0], subst))
```

```
    else:
        return None

def unify_var(var, x, subst):
    """Handle unification for variables."""
    if var in subst:
        return unify(subst[var], x, subst)
    elif x in subst:
        return unify(var, subst[x], subst)
    elif occurs_check(var, x, subst):
        return None
    else:
        subst[var] = x
        return subst

def occurs_check(var, x, subst):
    """Check for cyclic dependencies in substitution."""
    if var == x:
        return True
    elif isinstance(x, list):
        return any(occurs_check(var, xi, subst) for xi in x)
    elif x in subst:
        return occurs_check(var, subst[x], subst)
    return False

def resolution(kb, query):
    """Apply resolution algorithm to check satisfiability."""
    clauses = kb + [negate(query)]
    while True:
        new_clauses = []
        for i, ci in enumerate(clauses):
```

```
            for j, cj in enumerate(clauses):
                if i >= j:
                    continue
                resolvents = resolve(ci, cj)
                if [] in resolvents:
                    return True  # Query is satisfiable
                for res in resolvents:
                    if res not in new_clauses:
                        new_clauses.append(res)
        if all(new in clauses for new in new_clauses):
            return False  # Query is not satisfiable
        clauses.extend(new_clauses)

def negate(literal):
    """Negate a literal or a list of literals."""
    if isinstance(literal, list):
        return [negate(lit) for lit in literal]
    if literal.startswith("~"):
        return literal[1:]
    else:
        return f"~{literal}"

def resolve(clause1, clause2):
    """Resolve two clauses to find resolvents."""
    resolvents = []
    for lit1 in clause1:
        for lit2 in clause2:
            if lit1 == negate(lit2):
                new_clause = list(set(clause1 + clause2))
                new_clause.remove(lit1)
                new_clause.remove(lit2)
```

```
                if new_clause not in resolvents:
                    resolvents.append(new_clause)
    return resolvents
if __name__ == "__main__":
    # Example knowledge base and query
    knowledge_base = [
        ["~P", "Q"],
        ["P"],
        ["~Q", "R"],
        ["~R"]
    ]
    query = ["R"]  # Representing the query as a list
    print("Knowledge Base:", knowledge_base)
    print("Query:", query)
    result = resolution(knowledge_base, query)
    if result:
        print("The query is satisfiable.")
    else:
        print("The query is not satisfiable.")
```

Output:

```
Knowledge Base:
[
    ["~P", "Q"],
    ["P"],
    ["~Q", "R"],
    ["~R"]
]
Query:
["R"]
Knowledge Base: [['~P', 'Q'], ['P'], ['~Q', 'R'], ['~R']]
```

Query: ['R']

The query is not satisfiable.

Ex No: 2c

IMPLEMENTATION OF BACKWARD CHAINING

Aim:

To implement backward chaining.

Procedure:

Program:

```
# Facts: The knowledge base containing information about facts
facts = {
    'a': True,
    'b': True,
    'c': False
}
# Rules: List of rules where each rule is a tuple (conclusion, conditions)
rules = [
    ('d', ['a', 'b']),  # d can be concluded if a and b are true
    ('e', ['b', 'c']),  # e can be concluded if b and c are true
    ('f', ['d', 'e'])   # f can be concluded if d and e are true
]
# Backward Chaining Function
def backward_chaining(goal, facts, rules):
    # Check if the goal is already a fact
```